

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Data Preparation & Cleaning

Initial Exploration

- Imported dataset using pandas
- Checked structure with `df.info()`
- Generated summary statistics
- Identified 37 missing values in Review Rating

Feature Engineering

- Created age groups by binning
- Standardized columns to snake case
- Dropped redundant `promo_code_used`
- Integrated with DuckDB for SQL analysis





Revenue by Gender

\$157,890

Male Customers

Total revenue generated

\$75,191

Female Customers

Total revenue generated

Male customers generated over double the revenue of female customers in this dataset.

Product Performance Insights



Top Product

Jewelry leads with 171 orders in Accessories category



Top Category

Clothing drives highest revenue at ~\$100K with 171 orders



Most Discounted

Hats have highest discount rate at 50%, followed by Sneakers at 49.66%

Top Products by Rating

1

Gloves

Average rating: 3.86

2

Sandals

Average rating: 3.84

3

Boots

Average rating: 3.82

4

Hat

Average rating: 3.80

5

Skirt

Average rating: 3.78

Shipping & Subscription Analysis

Shipping Comparison

Standard: \$58.46 avg purchase

Express: \$60.48 avg purchase

Express shipping customers spend ~\$2 more per purchase

Subscription Status

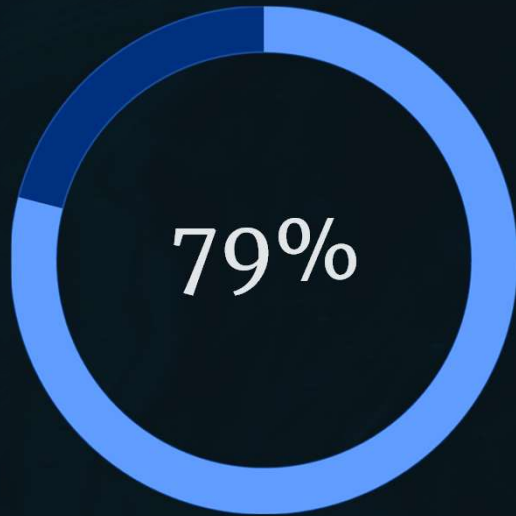
No: 2,847 customers, \$170,436 total revenue

Yes: 1,053 customers, \$62,645 total revenue

Non-subscribers generate 2.7x more revenue

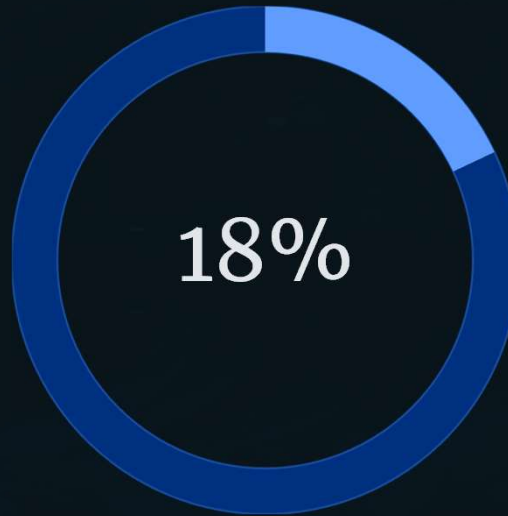


Customer Segmentation



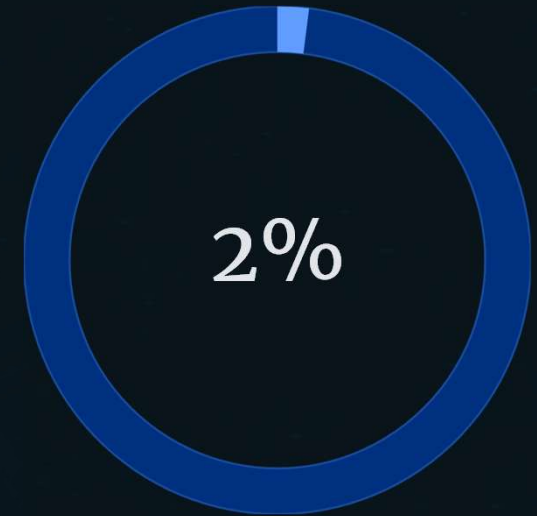
Loyal Customers

3,116 customers with established purchase history



Returning Customers

701 customers making repeat purchases

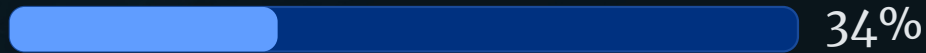


New Customers

83 customers with first-time purchases

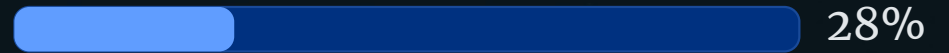
958 repeat buyers (over 5 purchases) have subscriptions, while 2,518 do not.

Revenue by Age Group



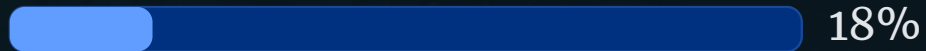
Middle-aged

\$89,445 revenue from 1,503 customers



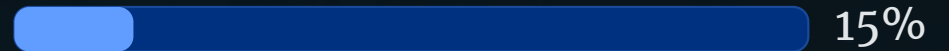
Adult

\$65,842 revenue from 1,103 customers



Senior

\$43,164 revenue from 723 customers



Young Adult

\$34,630 revenue from 571 customers

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to increase 27% subscription rate



Customer Loyalty Programs

Reward repeat buyers to move them into loyal segment



Review Discount Policy

Balance sales boosts with margin control, especially for high-discount products



Targeted Marketing

Focus on high-revenue age groups and express-shipping users

